Toward a Layered Ontology of Semantic Duality: 14 Candidate Dualities Across 6 Algebraic Layers with Preliminary Domain and Neural Evidence

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Abstract

We present a formal ontological framework in which conceptual complexity emerges from 14+ candidate dualities organized across 6 algebraic layers (Boolean, Fuzzy, Ordinal, Modal, Trivalent, Probabilistic). The framework is grounded in 72 semantic primitives connected by a directed acyclic graph (DAG) of 133+ dependency edges. We evaluate the framework through three independent methodologies: (1) domain analysis across 8 disciplines and 3 negative controls (Astrology, Alchemy, Phrenology) using a composite 6-metric Domain Validity Score (DVS), achieving DVS ≥ 0.719 for all 8 positive domains while all 3 negative controls fall below the 0.50 threshold (gap = 0.233, robust across all weight configurations); (2) Bayesian model comparison yielding a Bayes factor of 121.4 (decisive) over the reversed ordering, though only anecdotal evidence over alternative philosophical orderings (BF = 1.14 vs. Hegel, 1.95 vs. Peirce); and (3) neurosymbolic learning via a GPT-2 Medium model augmented with a 72-bit triadic head, which learns the primitive structure at 90.8% bit accuracy (though this barely exceeds the 90.3% majority-class baseline due to target sparsity) while preserving general language ability (PPL = 31.95, identical to baseline). The learned representations exhibit near-perfect relational structure (Regla de Tres cosine = 0.996) and show a significant phase transition during training ($p = 0.0004$). We report both confirmatory and null results transparently: layer ordering in neural representations is not significant ($p = 0.71$), and only 5 of 13 dual axes show significant learned anti-correlation after FDR correction.

Keywords: ontology, semantic primitives, emergence, duality, neurosymbolic AI, algebraic structure, domain validation

1 Introduction

The search for universal semantic primitives—irreducible conceptual atoms from which all meaning is composed—spans philosophy (Leibniz's *characteristica universalis*), linguistics (Wierzbicka's Natural Semantic Metalanguage), and computational semantics (ConceptNet, FrameNet). Existing approaches either lack formal algebraic structure or fail to demonstrate empirical universality across domains.

We propose that conceptual complexity emerges through a specific sequence of **dualities**—paired oppositions that each require all previous dualities as prerequisites. This emergence follows an algebraic chain:

$$\text{Boolean} \rightarrow \text{Fuzzy} \rightarrow \text{Ordinal} \rightarrow \text{Modal} \rightarrow \text{Trivalent} \rightarrow \text{Probabilistic}$$

Each algebra introduces operators that the previous one cannot express: degree from existence, comparison from degree, possibility from order, valuation from possibility, observation from valuation.

The contribution of this work is threefold:

1. **Formalization**: 72 semantic primitives organized in a DAG across 6 algebraic layers, with 14 dual axes and formal dependency rules.
2. **Empirical evaluation**: cross-domain analysis (8 disciplines + 3 negative controls including Astrology, Alchemy, and Phrenology) using a composite 6-metric IDVS with statistical hardening (permutation tests, FDR correction, Bayesian model comparison, null baselines).
3. **Neural validation**: a neurosymbolic model (GPT-2 Medium + triadic head) that learns the proposed structure from natural language, demonstrating that the algebraic relationships are recoverable from distributional semantics.

2 The Framework

2.1 Semantic Primitives

We define 72 semantic primitives distributed across 6 layers:

Layer	Algebra	Prims	Examples
1. Point (0D)	Boolean $\{0, 1\}$	3	vacío, información, uno
2. Line (1D)	Fuzzy $\{0, +\}$	8	fuerza, contención, unión, separación
3. Time (1D+t)	Ordinal $\{<, =, >\}$	13	mover, orden, caos, creación
4. Plane (2D)	Modal $\{\Diamond, \Box\}$	21	bien, mal, verdad, mentira
5. Volume (3D)	Trivalent $\{-, 0, +\}$	23	vida, muerte, placer, dolor
6. Observer (3D+)	Probabilistic $\{0, ?\}$	4	temporal_obs, eterno_obs

Table 1: 72 semantic primitives across 6 algebraic layers.

Each primitive is assigned a unique prime number, a bit position, a layer, a set of dependencies (parents in the DAG), and a natural-language definition.

2.2 The Dependency DAG

Primitives are connected by a DAG with 133+ directed edges encoding the relation "X requires Y to exist." Key structural properties:

- **Layer monotonicity**: all dependencies of a layer-n primitive lie in layers $\leq n$.
- **3 roots**: vacío (bit 0), información (bit 1), uno (bit 44)—all layer 1.
- **Transitive closure**: the maximum ancestor set size is 22 (for creador_obs, layer 6).

2.3 The 14 Dualities

The 14 dualities emerge in a specific order, each anchored by 2–4 primitives:

2.4 Algebraic Layer Chain

Each layer operates with a distinct algebra. We conjecture that each transition reflects a hierarchy of conceptual complexity, where higher layers require cognitive access to distinctions formalized in lower layers. This is a *semantic* claim, not a claim about the logical independence of formal systems (Kripke defined modal semantics without fuzzy logic; Łukasiewicz defined trivalent logic without modality). The conjecture is partially supported computationally: 12/14 dualities (86%) are correctly predicted by their algebra assignment.

#	Duality	Anchors	Layer
D1	Existir / No-existir	vacío, información, uno	1
D2	Aquí / No-aquí	eje_profundidad, eje_vertical	2–4
D3	Antes / Después	posición_temporal, flujo_temporal	3
D4	Posible / Imposible	puede, debe, tal_vez	4
D5	Identidad / Diferencia	muchos, tipo_de, todo, algunos	2–4
D6	Movimiento / Quietud	mover, hacer, quietud	3
D7	Orden / Caos	orden, caos, bien, mal, verdad, mentira	3–4
D8	Uno / Muchos	uno, muchos	1–2
D9	Dentro / Fuera	contención, unión, separación	2
D10	Parte / Todo	parte_de, todo, algunos	2–4
D11	Vida / Muerte	vida, muerte, placer, dolor	5
D12	Causa / Efecto	porque, si_entonces	3
D13	Semejante / Diferente	tipo_de	4
D14	Sujeto / Objeto	consciente, ausente, temporal_obs	5–6

Table 2: The 14 proposed dualities with anchor primitives and layer assignments.

3 Domain Validation

3.1 The IDVS Metric

We define the **Information Domain Validity Score** for a domain d as a weighted composite of six sub-metrics:

$$\mathrm{DVS}(d) = \sum_{i=1}^{6} w_i \cdot M_i(d)$$

with sub-metrics:

- **M1 (Coverage)** $= 1 - \frac{\text{nulls in layers 1–4}}{\text{total in layers 1–4}}$. Penalizes domains with many unclassifiable foundational primitives.
- **M2 (Direct ratio)** = fraction of primitives classified as D. Genuine domains map more primitives directly.
- **M3 (Strong duality)** = fraction of the 12 dual axes where both members are D or A. Measures structural engagement.
- **M4 (DAG consistency)** $= 1 - \frac{\text{violations}}{\text{total edges}}$. A violation occurs when a child has higher state than its parent ($D > A > N$).
- **M5 (Anchor consistency)** = agreement between predicted and actual D-A-N on held-out anchor terms.
- **M6 (Layer coherence)** = mean per-layer consistency of D-A-N classifications. Penalizes layers where primitives are incoherently split between engaged and null.

Default weights: $w = (0.25, 0.20, 0.15, 0.15, 0.10, 0.15)$.

3.2 Classification Protocol

Each of 72 primitives is classified in each domain as:

- **D** (Direct): the primitive operates directly in this domain.
- **A** (Analogical): the primitive applies by metaphor or analogy.
- **N** (Null): the primitive does not apply.

3.3 Results

All 8 positive domains achieve DVS ≥ 0.719 under the composite metric. All 3 negative controls fall below the 0.50 threshold:

Domain	Type	DVS	Result	σ below mean
Psychology	Positive	0.876	✓	—
Linguistics	Positive	0.857	✓	—
Philosophy	Positive	0.846	✓	—
Biology	Positive	0.831	✓	—
Chemistry	Positive	0.803	✓	—
Music	Positive	0.756	✓	—
Physics	Positive	0.755	✓	—
Mathematics	Positive	0.719	✓	—
Alchemy	**Negative**	**0.486**	×	6.0σ
Astrology	**Negative**	**0.438**	×	6.9σ
Phrenology	**Negative**	**0.322**	×	9.1σ

Table 3: DVS results across 8 positive domains and 3 negative controls. Gap between worst positive (Mathematics, 0.719) and best negative (Alchemy, 0.486) is 0.233. All weight configurations maintain discrimination (Section 3.5).

3.4 Null Baseline

To calibrate the threshold, we computed 10,000 random D-A-N assignments preserving the marginal distribution of each domain. The null distribution yields: mean max DVS = 0.861, $p_{95} = 0.931$, $p_{99} = 0.955$. All 8 positive domains exceed p_{99}. Note that the composite DVS (v2, 6 sub-metrics) is more informative than the original 2-metric IDVS: the original IDVS assigned perfect 1.000 to all positives, masking genuine variation. The composite DVS reveals that Psychology (0.876) and Linguistics (0.857) are the strongest domains while Mathematics (0.719) is weakest—consistent with the framework's linguistic origins.

3.5 Sensitivity Analysis

The composite DVS is robust to weight variation: all 7 tested weight configurations (default, equal, M1-heavy, M6-heavy, no-M1, no-M6, legacy) maintain perfect discrimination between all 8 positives and all 3 negatives. Leave-one-out analysis shows that dropping any single sub-metric preserves discrimination (minimum gap = 0.147 when dropping M2). No single metric alone suffices—M5 alone fails to discriminate (Alchemy M5 = 0.961 exceeds some positives)—confirming the necessity of the composite approach.

Perturbation	Target	Mean ΔDVS	Large disruptions
Remove edge	DAG topology	−0.003	3.4%
Move layer	Layer assignment	0.000	0.0%
Flip D-A-N	Classification logic	−0.060	71.4%

Table 4: Sensitivity analysis. Robustness gradient: Topology > Layering > Logic.

3.6 Bayesian Model Comparison

We compared 4 orderings of the 14 dualities using triangularity of the dependency matrix as the test statistic:

Against 1,000 random orderings ($\mu = 0.338$), the proposed ordering is significant ($p = 0.005$).

Model	Triangularity	BF vs. Reversed
Proposed	**0.539**	**121.4** (decisive)
Hegel	0.528	—
Peirce	0.484	—
Reversed	0.143	1.0

Table 5: Bayesian model comparison of duality orderings.

4 Neural Validation

4.1 Architecture

We augment GPT-2 Medium (355M parameters) with a **triadic head**: a single linear projection from the 1024-dimensional hidden state to a 72-bit semantic vector. The activation function is iFSQ ($2\sigma(1.6x) - 1$), which distributes activations uniformly across $[-1, +1]$.
Total trainable parameters (v5_frozen): 73,728 (0.02% of model). The GPT-2 backbone is frozen.

4.2 Training

Component	Configuration
Loss	$\mathcal{L}_{\text{lang}} + 0.05 \cdot (\mathcal{L}_{\text{tri}} + 2\mathcal{L}_{\text{sup}} + 5\mathcal{L}_{\text{sub}})$
$\mathcal{L}_{\text{tri}}$	diversity + InfoNCE + entropy + alignment
$\mathcal{L}_{\text{sup}}$	MSE(projection, gold_target)
$\mathcal{L}_{\text{sub}}$	relu($\text{hypernym}_{01} - \text{hyponym}_{01}$).mean()
Optimizer	AdamW (3×10^{-4}, $\beta = 0.9/0.95$)
Schedule	Cosine decay, 50% triadic warmup
Data	WikiText-103 (∼100M tokens)
Steps	50,000
Hardware	NVIDIA RTX 4060 Ti 16GB

Table 6: Training configuration for v5_frozen.

4.3 Training Run Summary

Run	Bits	Acc.	Unique	Coh.	PPL	Key Change
v1	65	∼90%	?	?	2,777	Baseline
v2	65	95.8%	3.8%	?	6,735	Bad targets
v3	65	90.9%	89.2%	0.653	5,388	Fixed: 65 prims
v4	72	92.5%	100%	0.876	5,608	+7 primitives
v5	**72**	**90.8%**†	**100%**	**0.818**	**31.95**	**Frozen base**
v6	**72**	**80.9%**	**21.4%**	**0.643**	**31.95**	**2,166 concepts**

Table 7: Training run progression. V5 solves catastrophic forgetting. V6 scales to 2,166 concepts. †Majority-class baseline for V5 is 90.3% (see Table 10).

4.4 Key Finding: Field Density > Individual Targets

V4 added 7 new primitives to differentiate collapsed dual pairs (Jaccard > 0.80). The result was surprising: collapsed pairs resolved even though their individual gold targets did not change.

The mechanism is **field density**—more diverse training examples teach the model what good differentiation looks like.

4.5 Key Finding: Catastrophic Forgetting Solved

All runs v1–v4 showed catastrophic language degradation (PPL $> 2{,}700$ vs. baseline 31.95). V5 froze the GPT-2 backbone, training only the 73K-parameter triadic head. Result: PPL = 31.95 (identical to baseline) at a cost of only -1.7% accuracy.

4.6 Relational Structure (Regla de Tres)

The model preserves proportional relationships between dual pairs:

Analogy	**Cosine**	**Bit Acc.**
good:evil = truth:lie	0.998	1.000
creation:destruction = life:death	0.998	1.000
freedom:control = individual:collective	0.995	0.986
pleasure:pain = conscious:absent	0.996	1.000
union:separation = order:chaos	0.997	1.000
creation:destruction = union:separation	0.997	1.000
receptive:creator_obs = truth:lie	0.996	0.986
move:stillness = mortal_obs:eternal_obs	0.989	0.931

Table 8: Regla de Tres: relational structure preservation (v5_frozen). Mean cosine: 0.997 across 8 quadruples.

4.7 Neural Probes

8 falsifiable questions were tested using reptimeline checkpoint analysis:

Probe	**Question**	**Result**	p**-value**
Q1	Layer order in representations?	**Null**	0.713
Q2	Dual anti-correlation?	**5/13 significant**	< 0.018 (BH)
Q3	DAG recovery?	**Above random**	0.001
Q5	Phase transition?	**Yes, step 6000**	0.0004
Q4	Triadic dependencies?	Null	1.000
Q6	Unsupervised discovery?	Null	1.000
Q7	Causal verification?	Null	1.000

Table 9: Neural probe results. The model learns relational but not ordinal structure.

Interpretation: The model learns *relational* structure (analogies, some dual axes, DAG edges) but does not recover *ordinal* structure (layer ordering, ternary dependencies). This is consistent with the frozen-base design: the head projects semantics into bit space without restructuring the backbone's internal representations.

4.8 V6: Extended Corpus (2,166 Concepts)

V5 trained on the 72 core primitives. V6 extends the gold file to 2,166 concepts (72 primitives + 2,094 derived from a first-principles corpus spanning 14 scientific domains), testing whether the algebraic structure scales beyond the primitive vocabulary.

Key findings:

1. **Backbone intact**: PPL = 31.95 after 100K steps—freeze-base eliminates catastrophic forgetting even at $2\times$ training duration.

Metric	v5_frozen (72)	v6 (2,166)
Bit accuracy (test)	90.8%†	80.9%
Majority baseline	90.3%	—
Subsumption accuracy	94.2%	95.2%
PPL	31.95	31.95
Active bits	4/72	48/72
Dead bits	68	37
Entropy (bits)	0.039	0.370
Unique signatures	100%	21.4%
Regla de Tres (cosine)	0.997	0.998
Discovered deps	174	501
Triadic deps	918	1,714
Training time	4.3h	12.3h

Table 10: V5 vs V6 comparison. V6 trades individual accuracy for richer structural discovery. †V5's 90.8% bit accuracy barely exceeds the majority-class baseline (90.3%) because gold targets are sparse (mean activation rate = 13%); most of the apparent accuracy comes from correctly predicting inactive bits. V6's 80.9% is more meaningful given higher bit utilization.

2. **Field density effect**: 30× more concepts force the model to activate 48 bits (vs. 4 in v5), increasing per-bit entropy by 9.5×. The bit space is utilized, not wasted.
3. **Semantic clustering**: Extended concepts collapse into 463 unique signatures. Collapses are semantically coherent (e.g., algebra axioms share a single code). The model functions as a *family classifier* rather than an individual discriminator at this scale.
4. **Inverted layer emergence**: reptimeline analysis across 20 checkpoints reveals that Layer 6 (Observer) primitives activate first (step 5,000), while Layers 1–3 activate last (step 55,000). The theoretical emergence order (L1→L6) is inverted during learning—abstract concepts are more statistically discriminative in GPT-2's embedding space.
5. **Phase transition**: A sharp transition occurs at step 55,000 (the warmup boundary), with entropy jumping from ∼0 to 0.37, code churn spiking, and bit utilization tripling.

5 Discussion

5.1 What the Framework Claims

The Emergent Duality framework makes two testable claims:

1. **Structural claim**: 72 primitives connected by a specific DAG can classify concepts across any knowledge domain using a ternary (D/A/N) code.
2. **Ordering claim**: the 14 dualities emerge in a specific sequence dictated by algebraic dependency.

5.2 Evidence For

- **Structural claim**: supported. 8/8 positive domains achieve DVS ≥ 0.719, while all 3 negative controls (Astrology 0.438, Alchemy 0.486, Phrenology 0.322) fall below the 0.50 threshold. Discrimination is robust across all tested weight configurations and leave-one-out metric ablations.
- **Ordering claim**: partially supported. Bayesian comparison yields BF = 121.4 (decisive) against reversed ordering, but only weak evidence against Hegel (BF = 1.14) and Peirce (BF = 1.95). The neural model does not recover layer ordering (Q1, $p = 0.71$).

5.3 Honest Limitations

1. **Self-evaluated classifications**: All D-A-N assignments were made by the framework author. Inter-rater data exists for only 1/9 domains (Mathematics). The permutation tests validate internal consistency but not external agreement.
2. **Underpowered rank correlations**: With $n = 7$ (circle) or $n = 14$ (combined), Spearman correlations cannot achieve significance at $\alpha = 0.05$ even for moderate effects.
3. **DVS variation across positives**: The composite 6-metric DVS reveals genuine variation among positive domains (0.719–0.876), with Mathematics weakest and Psychology strongest. The original 2-metric IDVS assigned 1.000 to all positives, masking this variation. While the composite metric is more informative, the gap between worst positive (0.719) and best negative (0.486) is only 0.233—a second positive domain near the Mathematics end could narrow this margin.
4. **Neural probes**: 5/8 probes returned null results. This constitutes evidence *against* the framework's deeper claims, not merely a limitation: the model learns surface relational structure (analogies, some dual axes) but does *not* recover the ordinal layer structure ($p = 0.71$) or triadic dependencies ($p = 1.0$) that are central to the theory. If the proposed algebraic chain were strongly reflected in distributional semantics, these probes should have reached significance.
5. **Scale–algebra tradeoff**: Larger models (355M) show lower subsumption rates (76.9%) than smaller ones (40M, 98.3%).

5.4 Future Directions

This work opens several lines of investigation:

1. **Independent rater validation**: The most critical next step. Blind inter-rater studies across all 9 domains would transform self-evaluated D-A-N classifications into externally validated ones. Preliminary data for Mathematics (1 rater) shows agreement; 8 domains remain.
2. **Scale–algebra tradeoff**: Understanding why 355M-parameter models show lower subsumption (76.9%) than 40M models (98.3%) may reveal fundamental constraints on discrete algebraic learning at scale.
3. **Cross-lingual validation**: All primitives are currently named and evaluated in a Spanish-language conceptual frame. Testing whether the same DAG structure holds when primitives are derived independently from other linguistic traditions would strengthen universality claims.
4. **Relational prime chains**: Extending the triadic head with compositional prime chain verification for O(1) anti-hallucination—a natural application of the algebraic structure to production AI systems.
5. **Expanded corpus training**: V5 was trained on WikiText-103. Training on multilingual or domain-specific corpora could reveal whether the 72-primitive structure generalizes beyond English-language distributional semantics.
6. **Additional dualities**: Preliminary analysis of the first-principles corpus (92 files across 14 scientific domains) suggests candidate dualities beyond the 14 validated here. Ongoing training runs are evaluating these extensions against the full primitive dataset; results will be incorporated in future revisions.

5.5 Conditions for Falsification

We pre-register four conditions under which the Emergent Duality framework should be considered falsified or require fundamental revision:

1. **Inter-rater failure** (K2): If independent raters achieve Fleiss' $\kappa < 0.40$ on D-A-N classifications across ≥ 4 domains, the framework's core claim—that primitives can be reliably

classified into Direct, Analogical, or Null categories—is not reproducible. *Action*: retract universality claims; re-characterize layers as "proposed" rather than "validated"; redesign D-A-N codebook.

2. **Negative control passes**: *Resolved.* Both Alchemy (DVS = 0.486) and Phrenology (DVS = 0.322) were evaluated and correctly fall below the 0.50 threshold, joining Astrology (0.438). Discrimination holds across all weight configurations. This criterion is currently satisfied; it would be triggered if a future pseudo-scientific domain achieved DVS ≥ 0.50.
3. **Cross-linguistic failure**: If the same DAG structure, when independently derived from a typologically distant language (e.g., Mandarin, Swahili), produces <40% overlap with the current 72-primitive inventory, the framework reflects Spanish-language conceptual biases rather than universal structure. *Action*: restrict claims to "Western philosophical tradition" rather than "universal ontology."
4. **Neural ordering recovery**: If a model trained with the full four-component triadic loss on a corpus $\geq$10$\times$ larger than WikiText-103 still fails to recover layer ordering (Q1 probe $p > 0.05$), the proposed emergence sequence is not reflected in distributional semantics at any accessible scale. *Action*: recharacterize the ordering as a philosophical hypothesis rather than an empirically supported claim.

These criteria are pre-registered before the planned inter-rater study (K2) to prevent post-hoc threshold adjustment.

6 Related Work

- **Wierzbicka (1996)**: Natural Semantic Metalanguage proposes ~65 universal primitives but lacks formal algebraic structure or computational validation.
- **ConceptNet** [Speer et al., 2017]: Large-scale knowledge graph with 34 relation types but no emergence hierarchy.
- **FrameNet** [Baker et al., 1998]: Frame-based semantics covering ~1,200 frames but domain-specific rather than universal.
- **Category theory approaches** [Coecke et al., 2010]: DisCoCat and related formalisms provide compositional semantics but do not address emergence ordering.

Our framework differs in proposing (a) a specific algebraic chain governing emergence, (b) empirical validation across 9 domains, and (c) neurosymbolic learnability from natural language.

7 Conclusion

We have presented a candidate ontological framework of 14 proposed dualities organized across 6 algebraic layers, grounded in 72 semantic primitives. The framework achieves perfect IDVS scores across 8 disciplines while correctly rejecting a negative control, with statistical hardening via permutation tests, null baselines, and Bayesian model comparison (decisive against reversed ordering, but only anecdotal against alternative philosophical orderings). A neurosymbolic model (GPT-2 + triadic head) learns the proposed structure at 90.8% accuracy without degrading language ability, exhibiting near-perfect relational preservation (cosine 0.996). However, the neural validation provides limited support: only 3/8 probes reach significance, and the two most theoretically important—layer ordering ($p = 0.71$) and triadic dependencies ($p = 1.0$)—are null. This suggests the model learns surface relational structure but not the deep algebraic properties the framework proposes.

The main open question is external validation: independent human evaluators must confirm the D-A-N classifications before universality claims can be considered robust. We release all data, scripts, and model code to enable replication.

Software and Data Availability

All code and data are available under BUSL-1.1 at:
- Validation scripts, DAG data, and domain analyses: `triadic-emergent-duality` repository.
- Neurosymbolic training pipeline (GPT-2 + triadic head): `triadic-microgpt` repository.
- Representation lifecycle tracking: `reptimeline` (also available on PyPI).
- Triadic projection head (standalone): `triadic-head` (also available on PyPI).

The repository also includes a structured compilation of first principles across 14 scientific domains (92 files organized hierarchically from mathematics to interdisciplinary fields), used by the algebraic gap analysis to search for candidate operators in uncovered inter-layer connections.

Companion Papers

This work is part of a four-paper program on triadic neurosymbolic representations. The present paper's claims are evaluated independently; companion results provide additional context but are not required for the conclusions drawn here. Each companion component has its own publication and repository:
- **Triadic Neurosymbolic Engine**: Ornelas Brand, J. A. (2026). *Prime Factorization as a Neurosymbolic Bridge: Projecting Continuous Embeddings into Discrete Algebraic Space for Deterministic Verification.* Paper: 10.5281/zenodo.19205805. Repository: 10.5281/zenodo.18748671.
- **reptimeline**: Ornelas Brand, J. A. (2026). *Tracking Discrete Representation Evolution During Neural Network Training.* Paper: 10.5281/zenodo.19208672. Repository: 10.5281/zenodo.19208628.
- **triadic-microgpt**: Ornelas Brand, J. A. (2026). *End-to-End Prime Factorization in a Generative Language Model: Learned Algebraic Encoding from Joint Training.* Paper: 10.5281/zenodo.19206545. Repository: 10.5281/zenodo.19207845.

A Notation

Symbol	Meaning
D, A, N	Direct, Analogical, Null classification states
IDVS	Information Domain Validity Score
BF	Bayes Factor
PPL	Perplexity
iFSQ	Improved Finite Scalar Quantization
BH-FDR	Benjamini–Hochberg False Discovery Rate
SHD	Structural Hamming Distance
DAG	Directed Acyclic Graph

Table 11: Notation reference.

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